Executive Summary
Template

Recommended Solution:

- A Random Forest churn prediction model was developed and evaluated using multiple performance metrics. The model achieved **high overall accuracy (≈90%)**, largely due to correctly predicting non-churn customers. However, the **recall for churn prediction is low (≈6%)**, indicating that many actual churn cases are not being detected. The **ROC-AUC score of ~0.66** shows the model has moderate ability to distinguish between churn and non-churn customers but still requires improvement for reliable churn detection.

```
Accuracy: 0.9041621029572837
Precision: 0.8076923076923077
Recall: 0.05737704918032787
F1 Score: 0.10714285714285715
ROC-AUC: 0.6576476124908122

Confusion Matrix:
 [[3281    5]
 [ 345   21]]

Classification Report:

              precision    recall  f1-score   support

           0       0.90      1.00      0.95      3286
           1       0.81      0.06      0.11       366

    accuracy                           0.90      3652
   macro avg       0.86      0.53      0.53      3652
weighted avg       0.90      0.90      0.86      3652
```

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Situation:

The SME division wanted to better understand which customers are most likely to churn so that proactive retention strategies can be implemented. A machine learning model was developed using historical customer data to predict churn risk and identify customers that may require early engagement.

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Complication:

Customer churn directly impacts revenue and long-term growth. Without predictive insights, retention efforts are reactive and often applied too late. Additionally, the dataset is highly imbalanced because most customers do not churn. This makes it difficult for the model to correctly identify churn cases, as it tends to favor predicting the majority non-churn class.

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Question:

How can we identify customers who are at risk of churning early enough to implement effective retention strategies and protect recurring revenue?

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Business Impact:

The current model provides a baseline understanding of customer churn patterns. However, because it misses many churn cases, relying on it alone could limit the effectiveness of targeted retention campaigns. Improving the model's ability to detect churn will help the SME division focus retention efforts on the customers who are truly at risk.

Next steps:

- Address class imbalance using techniques such as **SMOTE, class weighting, or resampling** to improve churn detection.
- Optimize model parameters or test additional models (e.g., Gradient Boosting, XGBoost) to improve recall and ROC-AUC.